Composition of Substances and Solutions

Foundations of Chemistry

Dr. Michael Evans

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Georgia Institute of Technology

Outline

1. Formula Mass and the Mole Concept
2. Determining Empirical and Molecular Formulas
3. Molarity
4. Other Units for Solution Concentrations

Formula Mass and the Mole Concept

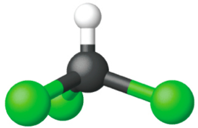
Molecular mass is a sum of atomic masses

Definition

Formula mass is the mass of a single molecule (molecular compounds) or functional unit (ionic or network compounds). Calculate it by summing all of the average atomic masses of atoms in the chemical formula.

Example

Element	Quantity		Average atomic mass (amu)		Subtotal (amu)
C	1	×	12.01	=	12.01
H	1	×	1.008	=	1.008
Cl	3	×	35.45	=	106.35
			Molecular mass		119.37

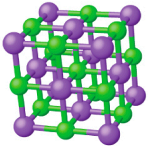


Ionic compounds have a formula mass

- Ionic compounds contain an infinite lattice of cations and anions (no ”molecular mass”)
- The chemical formula of an ionic compound refers to an electrically neutral “functional unit” of the lattice

Example

Element	Quantity		Average atomic mass (amu)		Subtotal
Na	1	×	22.99	=	22.99
Cl	1	×	35.45	=	35.45
			Formula mass		58.44



The mole is a convenient counting unit

Definition

The **mole** is a counting unit equivalent to 6.022×10^{23} (**Avogadro's number**) objects. It is defined as the number of carbon-12 atoms in 12.0 grams of the pure substance.

Definition

Molar mass is the mass of 1 mol of a substance. It is equal to the formula mass in amu.

Figure 1: One mole of S_8, $C_8H_{17}OH$, HgI_2, and CH_3OH (left to right).

We can use mass to count number of particles

Deriving Moles from Grams of a Compound

Our bodies synthesize protein from amino acids. One of these amino acids is glycine (molecular formula $C_2H_5O_2N$). How many moles of glycine molecules are contained in 28.35 g of glycine?

We can use mass to count number of particles

Deriving Grams from Moles of a Compound

Vitamin C is a covalent compound with the molecular formula $C_6H_8O_6$. The recommended daily dietary allowance of vitamin C for children aged 4 – 8 years is 1.42×10^{-4} mol. What is the mass of this allowance in grams?

We can use mass to count number of particles

Deriving Grams from Moles of a Compound

A packet of an artificial sweetener contains 40.0 mg of the molecular compound saccharin ($C_7H_5NO_3S$). How many saccharin molecules are in a 40.0-mg sample of saccharin? How many carbon atoms are in the same sample?

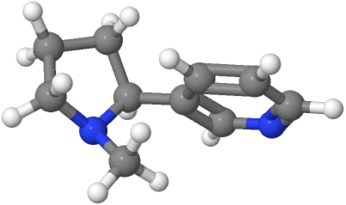
Determining Empirical and Molecular Formulas

Mass measurements give percent composition

Definition

Percent composition refers to the mass percents of the various elements in a compound. It is either:

A. Determined empirically through mass measurements, or
B. Derived from a given chemical formula



Elemental analysis: 60.33 g C, 10.95 g H, 11.73 g N, 66.98 g O

Percent composition can be derived from a formula

Deriving Percent Composition from Formula

What is the percent composition of ammonia (NH_3)? What is the percent composition of ammonium nitrate (NH_4NO_3)?

Empirical formula can be derived from percent composition

Deriving Empirical Formula from Percent Composition

1. Assume a basis of 100 g of substance.
2. Convert percent composition into moles of each element.
3. Divide by the smallest number of moles to determine ratios.
4. Scale ratios to obtain whole numbers for each element.

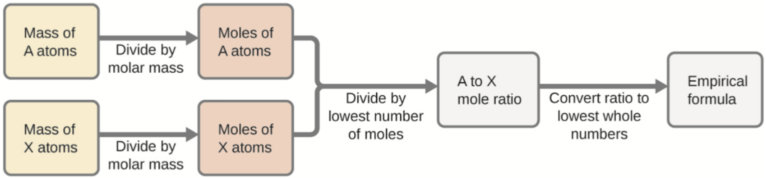


Figure 2: Determining empirical formula from mass measurements.

Empirical formula can be derived from mass measurements

Deriving Empirical Formula from Mass Measurements

A sample of the black mineral hematite, an oxide of iron found in many iron ores, contains 34.97 g of iron and 15.03 g of oxygen. What is the empirical formula of hematite?

Empirical formula can be derived from mass measurements

Deriving Empirical Formula from Percent Composition

The bacterial fermentation of grain to produce ethanol forms a gas with a percent composition of 27.29% C and 72.71% O. What is the empirical formula for this gas?

Molarity

Solutions are homogeneous mixtures

Solution A homogeneous mixture of two or more components

Solvent Component of a solution in greatest amount

Solute Components in much smaller amount

Concentration Relative amount or ”density” of a given component

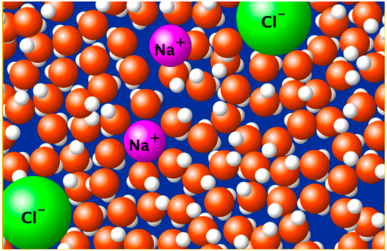
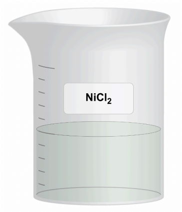


Figure 3: Submicroscopic view of a solution of sodium chloride in water.

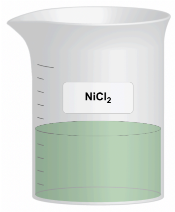
Concentration is the density of solute in a solution

Definition

The terms **dilute** and **concentrated** describe the relative amounts of solute in two or more solutions.



Dilute solution



Concentrated solution

A colored solute appears darker at higher concentrations. According to the **Beer-Lambert law**, the depth of color is linearly proportional to the molarity of a colored solute.

Molarity is the moles of solute per liter of solution

Definition

Molarity (M) is a measure of concentration defined as moles of solute per liter of solution (mol/L).

Molarities of an Ionic Solution

Ionic compounds dissociate into their component ions when dissolved in water. What are the molarities of Ni_2^+ and Cl^- in a 1.5 M solution of $NiCl_2$?

Dilution does not change the number of moles of solute

Definition

Dilution is the addition of solvent to a solution, which causes a decrease in concentration of all solutes.

- Moles of solute n remains constant during dilution: $n_1 = n_2 \Rightarrow M_1V_1 = M_2V_2$
- The **dilution equation** generalizes to any concentration units: $C_1V_1 = C_2V_2$

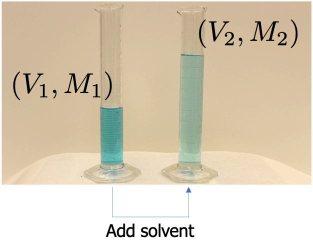


Figure 4: A stock solution (1) and diluted solution (2).

Concentration of a diluted solution can be calculated

Determining the Concentration of a Diluted Solution

If 0.850 L of a 5.00 M solution of $Cu(NO_3)_2$ is diluted to a volume of 1.80 L by the addition of water, what is the molarity of the diluted solution?

We can plan making a diluted solution

Volume of a Concentrated Solution for Dilution

What volume of stock 1.59 M KOH solution is required to prepare 5.00 L of 0.100 M KOH solution?

Other Units for Solution Concentrations

Concentration units combine mass, volume, and moles

Mass percentage Ratio of mass of solute to mass of solution

$(\%w/w)_i = \frac{m_i}{\sum_j m_j}$

Volume percentage Ratio of volume of solute to volume of solution

$(\%V/V)_i = \frac{V_i}{V_{tot}}$

Mass-volume percentage Ratio of mass of solute to volume of solution

$(\%w/V)_i = \frac{m_i}{V_{tot}}$

Mole fraction Ratio of moles of solute to total moles of all solution components

$x_i = \frac{n_i}{\sum_j n_j}$

Parts per million and parts per billion are very tiny units

”Parts per” Units

”Parts per million” refers to grams of a solute per million grams of solution, milliliters of solute per million milliliters of solution, etc. The units ppm and ppb are *very* small.

Parts per million (ppm) Concentration unit $\times 10^6$ million^{-1}

Parts per billion (ppb) Concentration unit $\times 10^9$ billion^{-1}

ppm and ppb are used in measures of water contaminants

Calculations Involving ppm and ppb

According to the EPA, when the concentration of lead in tap water reaches 15 ppb, certain remedial actions must be taken. What is this concentration in ppm? At this concentration, what mass of lead (μg) would be contained in a typical glass of water (300 mL)?